Software Development 2025-26
GE2b

Guided Exercise 2b
(Syntax & Structural Analysis)

Date: **23/04/2026**
TEAM: **02**

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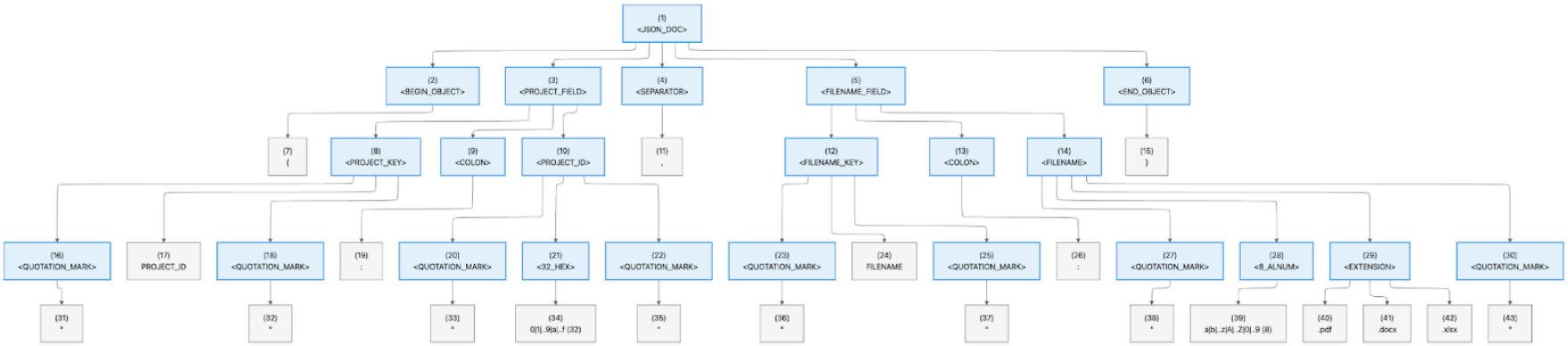
METHOD 2

Syntax Analysis- Grammar

```
<JSON_DOC> ::= <BEGIN_OBJECT><PROJECT_FIELD><SEPARATOR><FILENAME_FIELD><END_OBJECT>

<PROJECT_FIELD> ::= <PROJECT_KEY><COLON><PROJECT_ID>

<FILENAME_FIELD> ::= <FILENAME_KEY><COLON><FILENAME>

<PROJECT_KEY> ::= <QUOTATION_MARK> PROJECT_ID <QUOTATION_MARK>

<PROJECT_ID> ::= <QUOTATION_MARK><32_HEX><QUOTATION_MARK>

<FILENAME_KEY> ::= <QUOTATION_MARK> FILENAME <QUOTATION_MARK>

<FILENAME> ::= <QUOTATION_MARK><8_ALNUM><EXTENSION><QUOTATION_MARK>

<EXTENSION> ::= .pdf | .docx | .xlsx

<32_HEX> ::= 0|1|2|3|4|5|6|7|8|9|a|b|c|d|e|f (32)

<8_ALNUM> ::= a|...|z | A|...|Z | 0|...|9 (8)

<QUOTATION_MARK> ::= "

<BEGIN_OBJECT> ::= {

<END_OBJECT> ::= }

<SEPARATOR> ::= ,

<COLON> ::= :
```

Syntax Analysis - Derivation Tree



Non-Terminal Nodes: 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 13, 14, 16, 18, 20, 21, 22, 23, 25, 27, 28, 29, 30

Terminal Nodes: 7, 11, 15, 17, 19, 24, 26, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43

Syntax Analysis - Test Case Description

TC ID	Category	Description	Definition / Purpose	Expected Result
TC01	VALID	Valid JSON with .pdf extension	Verifies the system accepts a standard, well-formed JSON object using the .pdf extension.	Valid ProjectDocument Object
TC02	VALID	Valid JSON with .docx extension	Verifies the system accepts a standard, well-formed JSON object using the .docx extension.	Valid ProjectDocument Object
TC03	VALID	Valid JSON with .xlsx extension	Verifies the system accepts a standard, well-formed JSON object using the .xlsx extension.	Valid ProjectDocument Object
TC04	DELETION	Deletion of Node 1 - entire content	Validates that an empty file triggers a format exception.	EnterpriseManagementException("The file is not JSON formatted.")
TC05	DUPLICATION	Duplication of Node 1 - entire content	Ensures that two concatenated JSON objects are treated as invalid formatting.	EnterpriseManagementException("The file is not JSON formatted.")
TC06	DELETION	Deletion of Node 2/7 - the opening {	Validates that a missing root start-bracket fails JSON parsing.	EnterpriseManagementException("The file is not JSON formatted.")
TC07	DUPLICATION	Duplication of Node 2/7 - the opening {	Ensures that double-nesting without a matching close fails syntax validation.	EnterpriseManagementException("The file is not JSON formatted.")
TC08	MODIFICATION	Modification of Node 7 - the opening {	Verifies that replacing a curly bracket with another character (like [) is rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC09	DELETION	Deletion of Node 3 - project field	Ensures the parser fails when a leading comma exists without a preceding field.	EnterpriseManagementException("The file is not JSON formatted.")
TC10	DUPLICATION	Duplication of Node 3 - project field	Checks the system's resilience against duplicate keys or malformed key sequences.	EnterpriseManagementException("The file is not JSON formatted.")

TC11	DELETION	Deletion of Node 4/11 - separator ,	Verifies that missing property delimiters between fields cause a syntax failure.	EnterpriseManagementException("The file is not JSON formatted.")
TC12	DUPLICATION	Duplication of Node 4/11 - separator ,	Ensures that empty elements (double commas) are not allowed.	EnterpriseManagementException("The file is not JSON formatted.")
TC13	MODIFICATION	Modification of Node 11 - separator ,	Verifies that using an invalid separator (like .) results in a format exception.	EnterpriseManagementException("The file is not JSON formatted.")
TC14	DELETION	Deletion of Node 5 - filename field	Verifies that a key with a null/missing value at the end of the object is rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC15	DUPLICATION	Duplication of Node 5 - filename field	Verifies that duplicate field definitions at the structural level are rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC16	DELETION	Deletion of Node 6/15 - closing }	Checks that an unclosed JSON object triggers a parsing error.	EnterpriseManagementException("The file is not JSON formatted.")
TC17	DUPLICATION	Duplication of Node 6/15 - closing }	Verifies that trailing characters after the JSON object are rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC18	MODIFICATION	Modification of Node 15 - closing }	Verifies that an incorrect closing character (e.g.,]) is caught as a syntax error.	EnterpriseManagementException("The file is not JSON formatted.")
TC19	DELETION	Deletion of Node 8 - project key	Tests the behavior when a value exists without an associated key.	EnterpriseManagementException("The file is not JSON formatted.")
TC20	DUPLICATION	Duplication of Node 8 - project key	Checks for syntax errors caused by repeating the key name before the colon.	EnterpriseManagementException("The file is not JSON formatted.")
TC21	DELETION	Deletion of Node 9/19 - colon :	Verifies that a missing key-value separator is caught.	EnterpriseManagementException("The file is not JSON formatted.")
TC22	DUPLICATION	Duplication of Node 9/19 - colon :	Checks for syntax errors when multiple colons are present for one value.	EnterpriseManagementException("The file is not JSON formatted.")
TC23	MODIFICATION	Modification of Node 19 - colon :	Ensures that using a semicolon or other character as a separator is rejected.	EnterpriseManagementException("The file is not JSON formatted.")

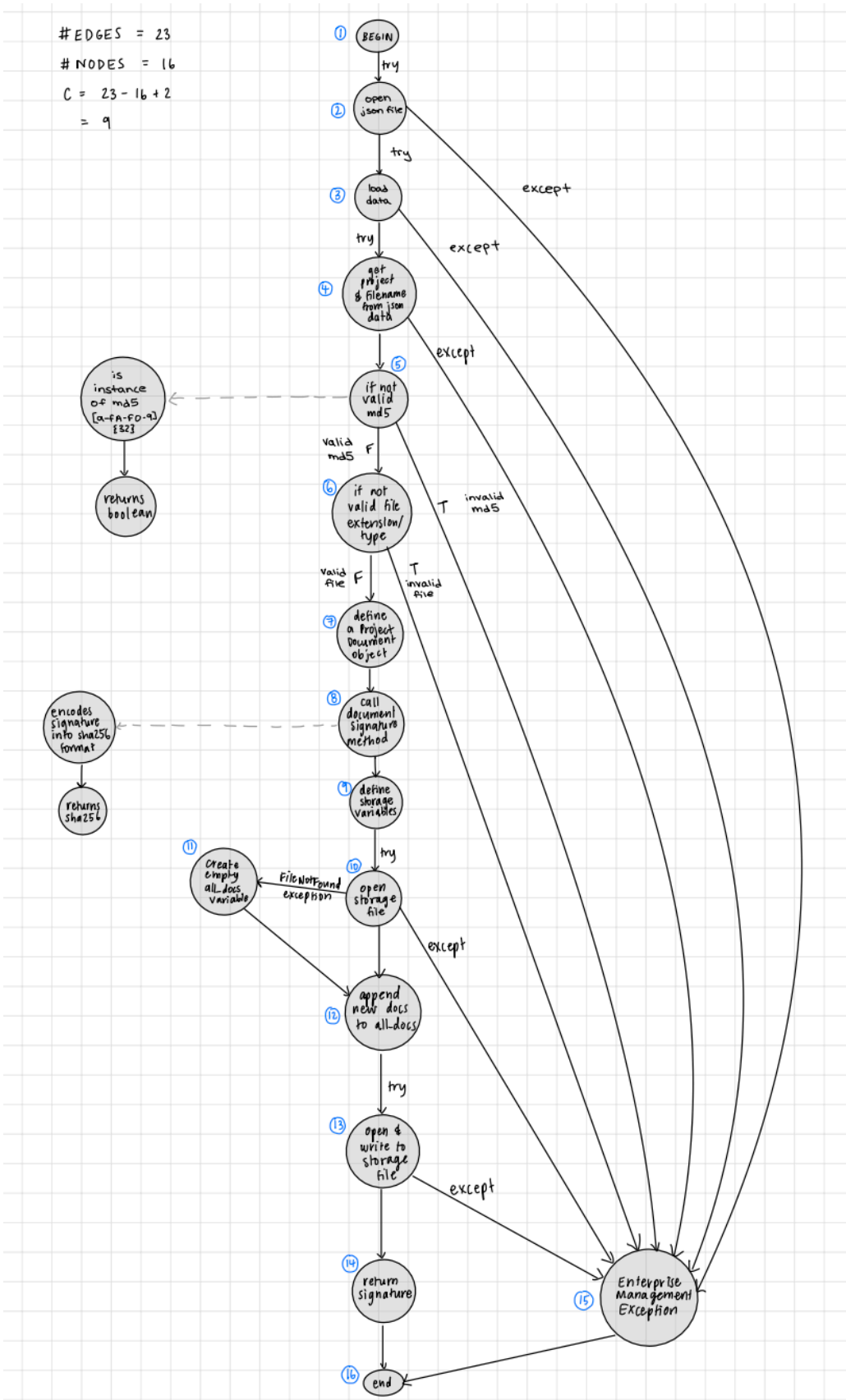
TC24	DELETION	Deletion of Node 10 - project id	Verifies that a key followed by a comma but no value is rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC25	DUPLICATION	Duplication of Node 10 - project id	Ensures that a value cannot be repeated without proper syntax.	EnterpriseManagementException("The file is not JSON formatted.")
TC26	DELETION	Deletion of Node 12 - filename key	Validates that a colon and value cannot exist without a preceding key.	EnterpriseManagementException("The file is not JSON formatted.")
TC27	DUPLICATION	Duplication of Node 12 - filename key	Checks for syntax errors when keys are duplicated before the colon.	EnterpriseManagementException("The file is not JSON formatted.")
TC28	DELETION	Deletion of Node 13/26 - colon :	Verifies the requirement of a colon between "FILENAME" and its value.	EnterpriseManagementException("The file is not JSON formatted.")
TC29	DUPLICATION	Duplication of Node 13/26 - colon :	Ensures double colons in the second field are rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC30	MODIFICATION	Modification of Node 26 - colon :	Rejects invalid characters (like ;) in the filename key-value pair.	EnterpriseManagementException("The file is not JSON formatted.")
TC31	DELETION	Deletion of Node 14 - filename value	Verifies that a key ending the object without a value is rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC32	DUPLICATION	Duplication of Node 14 - filename value	Ensures the parser rejects repeated value strings in a single pair.	EnterpriseManagementException("The file is not JSON formatted.")
TC33	DELETION	Deletion of Node 16/31 - quote "	Verifies that unquoted keys at the start of the object are rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC34	DUPLICATION	Duplication of Node 16/31 - quote "	Checks for syntax errors caused by empty or double-quoted key starts.	EnterpriseManagementException("The file is not JSON formatted.")
TC35	MODIFICATION	Modification of Node 31 - quote "	Validates that non-quote characters cannot start a key string.	EnterpriseManagementException("The file is not JSON formatted.")
TC36	DELETION	Deletion of Node 17 - PROJECT_ID content	Verifies that an empty key string is rejected by structural rules.	EnterpriseManagementException("JSON structure expected.")

TC37	DUPLICATION	Duplication of Node 17 - PROJECT_ID content	Checks for errors when the internal key text is doubled.	EnterpriseManagementException("JSON structure expected.")
TC38	MODIFICATION	Modification of Node 17 - PROJECT_ID content	Validates strict key naming (e.g., rejecting "PROJECT-ID").	EnterpriseManagementException("JSON structure expected.")
TC39	DELETION	Deletion of Node 18/32 - quote "	Checks that a key must have a closing quotation mark.	EnterpriseManagementException("The file is not JSON formatted.")
TC40	DUPLICATION	Duplication of Node 18/32 - quote "	Ensures trailing double quotes on a key are rejected.	EnterpriseManagementException("The file is not JSON formatted.")
TC41	MODIFICATION	Modification of Node 32 - quote "	Rejects keys ending with invalid characters instead of quotes.	EnterpriseManagementException("The file is not JSON formatted.")
TC42	DELETION	Deletion of Node 20/33 - quote "	Verifies that string values must start with a quotation mark.	EnterpriseManagementException("The file is not JSON formatted.")
TC43	DUPLICATION	Duplication of Node 20/33 - quote "	Ensures that value starts cannot have double quotes.	EnterpriseManagementException("The file is not JSON formatted.")
TC44	MODIFICATION	Modification of Node 33 - quote "	Rejects values starting with non-quote characters.	EnterpriseManagementException("The file is not JSON formatted.")
TC45	DELETION	Deletion of Node 21/34 - hex ID content	Checks if an empty string for Project ID is caught by logic rules.	EnterpriseManagementException("JSON has invalid values")
TC46	DUPLICATION	Duplication of Node 21/34 - hex ID content	Ensures Project ID length is strictly validated (rejecting double-length).	EnterpriseManagementException("JSON has invalid values")
TC47	MODIFICATION	Modification of Node 34 - hex ID content	Verifies that non-hexadecimal characters are rejected.	EnterpriseManagementException("JSON has invalid values")
TC48	DELETION	Deletion of Node 22/35 - quote "	Checks that a value must have a closing quotation mark.	EnterpriseManagementException("The file is not JSON formatted.")
TC49	DUPLICATION	Duplication of Node 22/35 - quote "	Ensures trailing double quotes on a value are rejected.	EnterpriseManagementException("The file is not JSON formatted.")

TC50	MODIFICATION	Modification of Node 35 - quote "	Rejects values ending with invalid characters instead of quotes.	EnterpriseManagementException("The file is not JSON formatted.")
TC51	DELETION	Deletion of Node 23/36 - quote "	Verifies the start-quote requirement for the second key.	EnterpriseManagementException("The file is not JSON formatted.")
TC52	DUPLICATION	Duplication of Node 23/36 - quote "	Ensures no double quotes at the start of the "FILENAME" key.	EnterpriseManagementException("The file is not JSON formatted.")
TC53	MODIFICATION	Modification of Node 36 - quote "	Rejects "FILENAME" key if it doesn't start with a quote.	EnterpriseManagementException("The file is not JSON formatted.")
TC54	DELETION	Deletion of Node 24 - FILENAME key text	Verifies that an empty filename key is rejected.	EnterpriseManagementException("JSON structure expected.")
TC55	DUPLICATION	Duplication of Node 24 - FILENAME key text	Checks for errors when the "FILENAME" key text is doubled.	EnterpriseManagementException("JSON structure expected.")
TC56	MODIFICATION	Modification of Node 24 - FILENAME key text	Ensures strict key naming (rejecting "FILE_NAME").	EnterpriseManagementException("JSON structure expected.")
TC57	DELETION	Deletion of Node 25/37 - quote "	Verifies that the "FILENAME" key must have a closing quote.	EnterpriseManagementException("The file is not JSON formatted.")
TC58	DUPLICATION	Duplication of Node 25/37 - quote "	Ensures no double quotes at the end of the "FILENAME" key.	EnterpriseManagementException("The file is not JSON formatted.")
TC59	MODIFICATION	Modification of Node 37 - quote "	Rejects "FILENAME" key if it ends with an invalid character.	EnterpriseManagementException("The file is not JSON formatted.")
TC60	DELETION	Deletion of Node 27/38 - quote "	Verifies that the filename value must start with a quote.	EnterpriseManagementException("The file is not JSON formatted.")
TC61	DUPLICATION	Duplication of Node 27/38 - quote "	Ensures no double quotes at the start of the filename value.	EnterpriseManagementException("The file is not JSON formatted.")
TC62	MODIFICATION	Modification of Node 38 - quote "	Rejects filename values starting with invalid characters.	EnterpriseManagementException("The file is not JSON formatted.")

TC63	DELETION	Deletion of node 28/39 - name prefix	Ensures that a file extension without a name prefix is invalid.	EnterpriseManagementException("JSON has invalid values")
TC64	DUPLICATION	Duplication of node 28/39 - name prefix	Checks for length validation of the filename prefix.	EnterpriseManagementException("JSON has invalid values")
TC65	MODIFICATION	Modification of node 39 - name prefix	Validates that special characters (e.g., `#`, `@`) are rejected.	EnterpriseManagementException("JSON has invalid values")
TC66	DELETION	Deletion of nodes 29/40/41/42 - extension	Ensures that a filename must have a supported extension.	EnterpriseManagementException("JSON has invalid values")
TC67	DUPLICATION	Duplication of node 29 - extension	Verifies that "double" extensions (e.g., `.pdf.docx`) are rejected.	EnterpriseManagementException("JSON has invalid values")
TC68	MODIFICATION	Modification of nodes 40/41/43 - extension type	Ensures that unsupported extensions (e.g., `.csv`) are rejected.	EnterpriseManagementException("JSON has invalid values")
TC69	DUPLICATION	Duplication of node 40 - .pdf	Specifically rejects doubled `.pdf` extensions.	EnterpriseManagementException("JSON has invalid values")
TC70	DUPLICATION	Duplication of node 41 - .docx	Specifically rejects doubled `.docx` extensions.	EnterpriseManagementException("JSON has invalid values")
TC71	DUPLICATION	Duplication of node 42 - .xlsx	Specifically rejects doubled `.xlsx` extensions.	EnterpriseManagementException("JSON has invalid values")
TC72	DELETION	Deletion of node 30/43 - quote "	Checks that the final value must have a closing quote.	EnterpriseManagementException("The file is not JSON formatted.")
TC73	DUPLICATION	Duplication of node 30/43 - quote "	Ensures no double quotes at the very end of the final value.	EnterpriseManagementException("The file is not JSON formatted.")
TC74	MODIFICATION	Modification of node 43 - quote "	Rejects final values ending with invalid characters.	EnterpriseManagementException("The file is not JSON formatted.")

Structural Analysis - Flow Control Graph



Structural Analysis - Paths

#	PATH	ID TEST	DESCRIPTION	EXPECTED RESULT
1	1_2_3_4_5_6_7_8_9_10_12_13_14_16	TC01	Happy path - every node is successful	update all_documents.json with data and return signature
2	1_2_15_16	TC02	Exception while opening json	EnterpriseManagementException("Input file not found.")
3	1_2_3_15_16	TC03	Exception while loading json	EnterpriseManagementException("The file is not JSON formatted.")
4	1_2_3_4_15_16	TC04	Exception while parsing data from json	EnterpriseManagementException("JSON does not have the expected structure.")
5	1_2_3_4_5_15_16	TC05	Invalid md5	EnterpriseManagementException("JSON has invalid values")
6	1_2_3_4_5_6_15_16	TC06	Invalid file extension type	EnterpriseManagementException("JSON has invalid values")
7	1_2_3_4_5_6_7_8_9_10_15_16	TC07	Exception while opening storage file to read	EnterpriseManagementException("Input file not found.")
8	1_2_3_4_5_6_7_8_9_10_11_12_13_14_16	TC08	Create a new file after FileNotFound exception	update all_documents.json with data and return signature
9	1_2_3_4_5_6_7_8_9_10_11_12_13_15_16	TC09	Exception when opening storage file to write	EnterpriseManagementException("Input file not found.")